

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech.

Branch: EXTC Engineering

Semester: V

Subject Code & Name: BTETOE505D Project Management and Operation Research

Max Marks: 60

Date: 19/08/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is Operations Research? Provide a concise definition and highlight its main objectives.	Understand, Remember	6
B) What are the limitations of Operations Research in dealing with real-world problems? Provide examples.	Understand, Remember	6
C) Explain the steps involved in formulating a linear programming problem. Provide an example to illustrate the process	Understand, Remember	6
Q.2 Solve Any Two of the following.		12
A) Define the dual of an LP problem. How is the dual problem related to the primal problem?	Understand	6
B) How is the graphical method used to solve an LP problem? Outline the general steps.	Understand	6
C) Explain the simplex algorithm for solving LP problems. Highlight the concept of basic feasible solutions and pivoting.	Understand, Remember	6
Q. 3 Solve Any Two of the following.		12
A) Explain the importance of determining the optimal replacement time for deteriorating items	Understand	6
B) Define a transshipment problem and Describe the process of formulating a transshipment problem	Understand, Remember	6
C) Explain how degeneracy can affect the solution process in transportation problems.	Understand	6
Q.4 Solve Any Two of the following.		12
A) Provide an example scenario where a single-server model might be suitable, and another scenario where a multiserver model is more appropriate.	Understand	6
B) Describe the Maximum Principle in the context of two-person zero-sum games.	Remember	6
C) Differentiate between pure strategies and mixed strategies in game theory.	Understand, Remember	6

Q. 5	Solve Any Two of the following.		12
A)	Explain PERT/CPM & its importance in project management	Understand,	6
B)	Why is a network diagram important in project management?	Understand,	6
C)	Explain the terminologies used in network diagram: Activity, Event, Critical path, Total project time , Duration , Float/slack	Understand, Remember	6

***** End *****